Does a streaming video-LLM know what it has forgotten?

Research briefing — two experiments, one probe validation, one collision check

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Qwen3-VL-8B-Instruct on EgoLifeQA · 373 questions · 5,540 + 17,311 scored records

Agenda

1. **Results: both experiments are in, and they disagree with the proposal.**
   The predicted signature is absent in both regimes. What replaced it is sharper.
2. **Confidence probe: audited and validated.**
   One sensitivity found; it is a result, not a nuisance.
3. **Related work: EgoSAT appeared in June.**
   It costs one novelty claim and independently replicates the central finding.
4. **Next steps and open decisions.**

Part 1

What the two experiments say

Two regimes, same backbone, same questions.
the prediction failed; what replaced it is a better claim.

Design summary

Same backbone, same questions, different memory architecture

	how context reaches the model	scored records
E1	re-encodes the whole window at every checkpoint	5,540
E2	one ingestion into a bounded KV cache, queried at checkpoints	17,311

Same frozen Qwen3-VL-8B, same 373 EgoLifeQA questions, same frame bank, same analysis code. EgoStream is a plug-in memory manager over the same weights — no bridge condition needed.

Accuracy: argmax vs gold. **Confidence**: max softmax({A,B,C,D}) at the first generated position; no text generated. Every Δ is a **paired within-question contrast**, cluster-bootstrapped over questions (5,000 draws).

The predicted signature is absent in both regimes

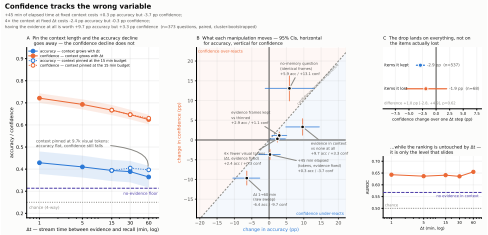
The recommendation is to retire the original hypothesis

Prediction: the gap between confidence and accuracy *widens* as evidence ages.

overconfidence gap, Δt $1 \rightarrow 60$ min	change (pp)	p
E1 — full-reprocess	-3.2 $[-7.5, +1.0]$	0.138
E2 — streaming KV cache	-4.0 $[-9.0, +1.0]$	0.105

Both point mildly toward *narrowing*. **E2** was the setting the hypothesis most deserved — a bounded cache actually destroys the evidence — and it still did not appear.

E1 (full-reprocess): confidence tracks the wrong variable



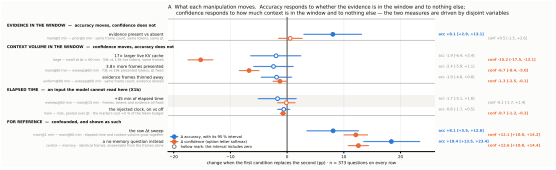
manipulation	Δacc	Δconf
+45 min elapsed *(tokens fixed)*	+0.3 [−2.9, +3.2]	−3.7 [−4.9, −2.4]
4× fewer tokens *(Δt fixed)*	+2.4 [+0.3, +4.7]	+0.3 [−0.5, +1.1]
evidence vs none *(token-matched)*	+9.7 [+4.8, +14.5]	+3.3 [+1.2, +5.4]

Confidence moves on elapsed time, which costs *no* accuracy. **Accuracy** moves on context length and evidence, which confidence barely registers.

Source: `figures/headline.pdf` (M7). Nothing redrawn.

E2 (streaming KV cache): the same dissociation, rotated

Accuracy responds to evidence; confidence responds to context volume



Elapsed-time axis is *unavailable*, not null — see backup.

Budget dose–response at $\Delta t = 60$ min:

Δt (min)	Δacc	Δconf
1	+3.0 [+0.0, +5.9]	−2.0 [−3.4, −0.6]
5	−0.0 [−3.8, +3.8]	−4.2 [−5.9, −2.4]
15	+1.6 [−2.1, +5.4]	−8.1 [−10.0, −6.2]
30	−0.0 [−4.3, +4.3]	−10.5 [−12.4, −8.5]
60	−1.9 [−6.4, +2.4]	−15.2 [−17.5, −13.1]

A 17× live-token budget is worth **−1.9 [−6.4, +2.4]** of accuracy and **−15.2 [−17.5, −13.1]** of confidence. Monotone on one measure, flat on the other.

Source: `figures/streaming_headline.pdf` (X10). Full figure in backup.

Cross-regime: the dissociation replicates and its driver rotates

Differenced contrasts — same manipulation in both regimes, paired on the same questions

manipulation, Δconfidence	**E1**	**E2**	difference (**E2** − **E1**)
evidence in context vs none	+3.3	+0.5	−2.8 [−5.1, −0.4]
+45 min elapsed, frames + evidence fixed	−3.7	−0.1	+3.5 [+1.7, +5.4]
3.8× more presented tokens	−0.6	−6.7	−6.1 [−8.1, −4.1]
no-memory vs recall question	+14.3	+12.6	−1.7 [−3.3, −0.1]

Invariant. Accuracy responds to evidence presence; confidence does not — in either regime.

Rotated. In **E1** confidence slid with elapsed time; in **E2** it slides with context volume. Both differences exclude zero.

Unified account and implications

Confidence is computed from surface properties of the context — how much visual input was processed (retained or not), and how long ago it says that was — **not from whether the needed information is present.** Remove one surface cue and it latches onto another. What it never latches onto is evidence.

What this does to the original claim:

- **Half 1** ("widening gap as evidence ages") — rejected twice.
- **Half 2** ("Δt-conditioned recalibration recovers reliability") — needs revision. Δt is the wrong conditioner; it worked in **E1** by accident. The variable that actually drives confidence is `live_tokens`.
- **Headroom differs by regime.** AUROC is flat over the **E1** sweep ($+0.012$ [-0.046, $+0.071$]) but decays in **E2** (-0.073 [-0.143, -0.003], $p = 0.039$). Recalibrating a level only works if the ranking survives.

The cheap fix looks cheaper the less realistic the memory system is — only visible because both regimes were run.

Part 2

Confidence probe validation

The alternative denominator was computed on the committed records. one sensitivity found; it is a result, not a nuisance.

Verdict: the restricted softmax holds

Vocabulary-normalised alternative changes nothing — except one number, for the wrong reason

Mean mass on {A,B,C,D}: 0.99988 (**E1**), 0.99546 (**E2**). Argmax disagreements: **0 of 24,433** records. Accuracy is identical under both definitions.

AUROC is invariant to how the four-way distribution is collapsed:

discriminator	E1	E2
max-p, restricted	0.6276	0.6196
max-p, vocab-normalised	0.6275	0.6198
negative entropy	0.6228	0.6209
margin (top1 − top2)	0.6254	0.6169

The one sensitivity: under vocabulary normalisation, **E2**'s evidence contrast flips from $+0.55$ $[-1.52, +2.65]$ to $+2.61$ $[+0.33, +4.94]$ — rewriting the "dissociation replicates" verdict. The flip rides on a small, heavy-tailed minority (130 rows (1.29%), over 79 of 373 questions with $>10\,\%$ off-letter mass, specific to the bounded cache).

The fix is to make the sensitivity visible: assert the denominator in the module and emit both values alongside the verdict.

Source: `memo/CONFIDENCE-PROBE.md`

Two by-products worth reporting

Implicit abstention. Off-letter mass is a metacognitive signal living where max-p is silent:

contrast	**E2** Δoff-letter
evidence removed	-3.01 $[-4.40, -1.82]$
raw Δt sweep	$+0.60$ $[+0.21, +1.11]$

The phenomenon does not exist in **E1** (max off-letter mass 0.0034). It converts one of **E2**'s nulls into a measured effect — at zero GPU cost.

Option-letter bias. Gold: A .222 B .256 C .261 D .261. Predicted: A .124 B .280 **C .435** D .161 (**E1**), A .134 **B .342** C .330 D .194 (**E2**). With no evidence the model predicts the same skewed letters at 0.659 confidence.

The paired design survives: the letter prior is a property of the question, fixed across arms, and differences out to first order. But EgoSAT already runs the check this project does not (option-order re-shuffle).

A **cyclic permutation arm** at one Δt bin would close this gap at $4\times$ cost on a subset.

Part 3

EgoSAT: collision and cross-validation

Prior art, not a scoop. It costs one novelty claim and independently replicates the central finding.

What EgoSAT is, and where it collides

Lei et al., arXiv 2606.24422, 23 Jun 2026

A streaming egocentric benchmark: 1,997 Ego4D sessions, 165 h, ∼4,800 MCQ, six tasks spanning retrospective / present / prospective, plus an answerability model and a training-free KV selection method.

Collision surface:

- The confidence probe (their Eq. 6 = our definition verbatim).
- Confidence vs temporal distance (same estimand, 8 / 16 / 24 s vs 1–60 min).
- Answerability-conditioned confidence ≈ our evidence contrast.
- Online vs offline (our regime pair, but cross-model not within-backbone).

Must fix in framing

- Drop "first to evaluate confidence in egocentric streaming."
- Attribute the probe to Tsvilodub et al. 2024 / EgoSAT Eq. 6.
- Retire "confidently wrong" as phrasing.

Position instead on **evidence age** and the **elapsed-time-vs-context-volume decomposition**.

Confidence vs temporal distance: their Table 2 and our Δt sweep

Their conf slope is fitted on multi-step retrospection at 8 / 16 / 24 s; ours spans 1–60 min

Their finding (Table 2, retrospection): six of eight ≥ 0 confidence slopes are flat or positive. Expected: negative. Their conclusion: "confidence does not reliably track temporal difficulty."

Against E2: consistent but uninformative. Our streaming elapsed-time contrast is flat (**−0.1 [−1.7, +1.4]**); their streaming models give +0.99 and +4.61. But their 16 s lag in ∼300 s sessions means evidence either survived the full session's eviction pressure or was dropped before the first probe — neither evidence state nor token count changes between the three probes. The flat slope is the expected null. Our **E2** carries the finding: 45 min, evidence controlled, within-backbone.

Against E1: apparent disagreement. Our full-reprocess elapsed-time contrast is **−3.7 [−4.9, −2.4]** — a clear decline, where their offline models are flat.

Three estimand differences that dissolve the E1 gap:

1. 45 min of elapsed time vs 16 s — 170× the horizon.
2. Their slope is fitted only on items where **all three probes are correct** — a correctness-selected subsample that truncates the range and selects high-confidence items; ours is unconditional over all 373 questions.
3. Their point estimates carry no intervals; half the table sits within rounding distance of zero.

Against **E1**: a difference in estimand, not a contradiction — but the first thing a reviewer will cite, so name Table 2 and state all three differences. Against **E2**: too short-ranged to count as independent support.

The central finding replicates outside this lab

Their answerability split, our evidence split — different dataset, different backbones, same shape

model / experiment	Δaccuracy	Δconfidence	under-reaction
Qwen2.5-VL-72B	+9.45 / −4.57		sign-flipped
Qwen2.5-VL-7B	+10.05 / +1.75		5.7×
TimeChat-Online-7B *(streaming)*	+4.51 / +0.80		5.6×
ROI-TimeChat-Online *(streaming)*	+5.42 / +0.74		7.3×
ours — E1	**+9.7 [+4.8, +14.5]**	**+3.3 [+1.2, +5.4]**	2.9×
ours — E2	**+8.1 [+2.9, +13.1]**	**+0.5 [−1.5, +2.6]**	CI through zero

No longer a one-dataset, one-model result. Cite as convergent evidence. Remaining differentiators: the token-matched decomposition, the within-backbone regime pair, the time scale (40–450×), and the statistics.

Part 4

Proposed next steps

Proposed follow-ups ranked by what they settle per GPU-hour.

Ranked follow-ups and open decisions

experiment	cost	what it settles
Strip timestamps from **E1**, re-score token-matched cells	≈1 GPU-h	Whether **E1**'s elapsed-time effect is a response to printed time tokens at all.
Recalibrate on `live_tokens`, per regime	free	The corrected Half-2 claim. AUROC asymmetry predicts success in **E1**, partial failure in **E2**.
Off-letter mass as a reported quantity	free, CPU	Turns the abstention signal into a result; surfaces the denominator sensitivity.
Cyclic option-permutation arm, one Δt bin	4×, one bin	The option-letter bias check EgoSAT runs and this project does not.
Explicit abstention (a fifth option)	∼2.4 GPU-h	Given the words to say "not enough information", does the model use them?

Open decisions:

1. Retire the original hypothesis and lead with the dissociation?
2. Which follow-up first? (Proposed order: timestamps-off, then recalibration.)
3. How hard to position against EgoSAT — convergent evidence vs competing framing?
4. Is the abstention channel worth a section, given it rides on a 1.4 % tail?

Part 5

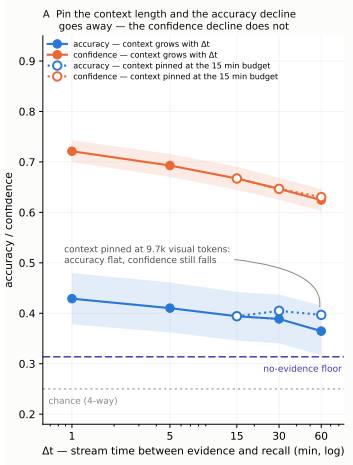
Backup

Raw material behind the claims.

Backup — E1, the raw Δt sweep

Δt (min)	acc %	conf %	gap	visual tokens
1	42.9	72.1	+29.2	1,279
5	41.0	69.3	+28.3	3,679
15	39.4	66.7	+27.3	9,679
30	38.9	64.7	+25.8	18,679
60	36.5	62.4	+26.0	36,679

Bin-1 accuracy 42.9 % sits 11.5 pp above the no-evidence floor of 31.4 %. Any statement about an accuracy *level* is made inside that narrow band.

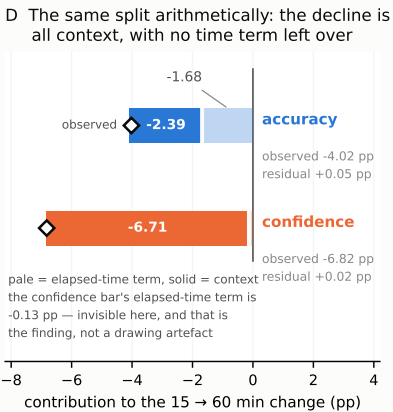


Backup — the decomposition, in both regimes

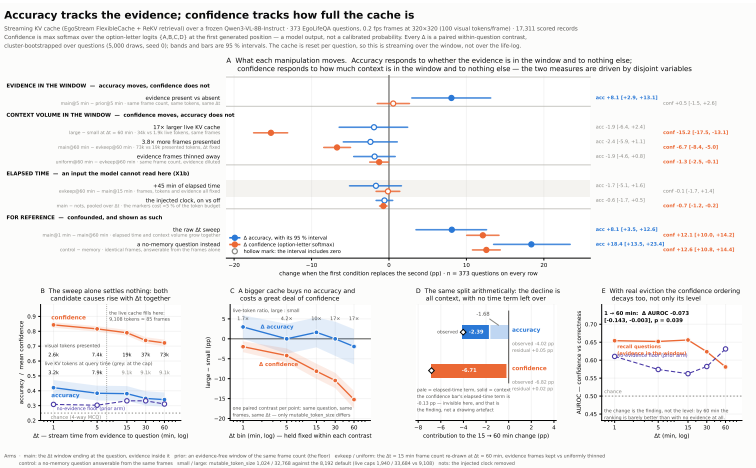
E1: Δt 15 → 60 min	acc	conf
elapsed-time term	+0.3 pp	−3.7 pp
context-length term	−2.4 pp	−0.3 pp

Accuracy's decline is entirely the context term; confidence's is almost entirely elapsed time.

E2: **−6.71 pp** of context against **−0.13 pp** of elapsed time, observed −6.82 pp, residual +0.02 pp.



Backup — E2, the whole headline figure



Backup — E2, full streaming contrasts

manipulation	Δaccuracy	Δconfidence
evidence in context vs none	+8.1 [+2.9, +13.1]	+0.5 [−1.5, +2.6]
+45 min elapsed, frames and evidence fixed	−1.7 [−5.1, +1.6]	−0.1 [−1.7, +1.4]
3.8× more presented tokens	−2.4 [−5.9, +1.1]	−6.7 [−8.4, −5.0]
3.7× live-token budget (offload-free pair, 60 min)	*null in every bin*	−6.0 [−7.3, −4.7]
17× live-token budget (spans the offload boundary)	−1.9 [−6.4, +2.4]	−15.2 [−17.5, −13.1]
evidence frames thinned away	−1.9 [−4.6, +0.8]	−1.3 [−2.5, −0.1]
injected clock on vs off	−0.6 [−1.7, +0.5]	−0.7 [−1.2, −0.2]
raw Δt sweep *(confounded)*	+8.1 [+3.5, +12.6]	+12.1 [+10.0, +14.2]
no-memory vs recall *(between-items)*	+18.4 [+13.5, +23.4]	+12.6 [+10.8, +14.4]

Source: `artifacts/x9_streaming.md`, generated from `x9_regime_comparison.json`.

Backup — three readings to rule out

1. **"The elapsed-time effect disappeared under streaming."** No. EgoStream's image path carries no time markers; the injected clock is unreadable once the cache saturates. The axis is **unavailable**, not null.
2. **"Streaming makes the model more overconfident."** Not supported — that is a level comparison, and the regime changes the system prompt and ingestion path. Contrasts are comparable; levels are not.
3. **"The model tracks the items it actually lost."** Not yet. **E2**'s kept-vs-lost difference is -3.8 $[-8.5, +1.1]$, $p = 0.118$, on 107 lost items — suggestive, underpowered.

Calling an **unavailable** axis a **null** is the most damaging misreading here — a reviewer will make it if the paper does not pre-empt it.

Backup — where EgoSAT appears to contradict, and why it does not

1. Confidence does not fall with distance in their table (six of eight ≥ 0 slopes).

Against **E2**: consistent but uninformative — neither evidence state nor token count changes between their three probes, so their flat slope is the expected null, not a replication. Against **E1**: apparent disagreement, for three estimand differences (45 min vs 16 s; slope fitted on all-correct items only; no intervals).

2. Their accuracy declines with distance (-2.6 to -7.4 pp over $8 \rightarrow 24$ s) **where our token-matched arms say elapsed time costs no accuracy.**

Their lag axis is uncontrolled for context volume. Their streaming models show the smallest or non-monotone declines — matching our account.

The defensible framing: their uncontrolled lag axis reproduces our uncontrolled raw sweep — and this project has the control that separates the components.

Backup — what the cache actually holds

The cache: pinned priming region, evictable mutable region (the budget knob), protected hat (last two frames). On overflow, the pruning pipeline drops the lowest-value and most redundant tokens.

Eviction is content-scored and global, not recency-ordered. "Older evidence is pushed out first" is not this policy. Eviction is permanent in the primary condition: 0 of 13,636 default/large rows offloaded anything, against 3,675 of 3,675 small rows.

The sweep crosses saturation at ≈85 frames (≈7 min). By $\Delta t = 60$ min only ≈11% of the presented tokens survive, live tokens pinned at 9,108 from 15 min on.

E1's clock is *text* the model tokenizes (<150.0 seconds>), not a positional encoding — confirmed by decoding the prompt through **E1**'s own input builder.

Source: `memo/STREAMING-CACHE.md` §2, §6–§7.

Backup — dataset curation: 500 → 373 questions

step	in	out	reason
EgoLifeQA A1 (full)	500	—	
Drop TaskMaster	500	437	prospective
Screen HabitInsight	437	376	no single anchor
Evidence span >15 min	376	373	ambiguous Δt

What survives. EntityLog / EventRecall / RelationMap at 125 / 124 / 124. Answer letters 83 : 96 : 97 : 97. All 373 appear in every Δt bin and every arm; no per-bin attrition.

Token matching. Matched arms hold visual tokens exactly at the $\Delta t = 15$ min level (mean 9,679) across all bins. Prior arms frame-matched to `main` with zero difference.

Flagged, not dropped. `need_audio` = true on 110/373 (29.5 %); results reported with and without.

Why the drops are defensible.

TaskMaster (63): questions about future plans — incompatible with a retrospective Δt sweep. 93.7 % require audio; the vision-only model is near-blind on them independently.

HabitInsight (61): habitual patterns with no single evidence event to anchor Δt. Scored separately, not dropped.

Span exclusions (3): q482 spans 50.7 h (3 evidence points across two days; alone 13 % of frame-encodes). q189/q346 at 19/21 min. Next-closest is 11.6 min — clean gap below the 15 min threshold.

96.5 % have a single evidence point (span = 0). The span filter affects a small tail; "Δt since evidence" is unambiguous for the rest.

Source: M1 `m1_questions.py`, M4 `m4_plan.py`; verified from `m1_questions.jsonl` and `m4_eval_plan.jsonl`.

Backup — how this deck is built

module	artifact	checks
B1 `b1_numbers.py`	`generated/numbers.tex`	Re-derives all macros from metrics JSON; memo-sourced values carry a literal check.
B2 `b2_figures.py`	`figures/*.pdf`	Panels are crops; manifest pins source figures by sha256.
B5 `b5_verify.py`	the built PDF	Nine checks: stale generators, missing figures, undefined macros, LaTeX errors, overfull slides.

No number on any slide was typed by hand. If an experiment is re-run, `make` rebuilds the deck and `make verify` fails until it is rebuilt.